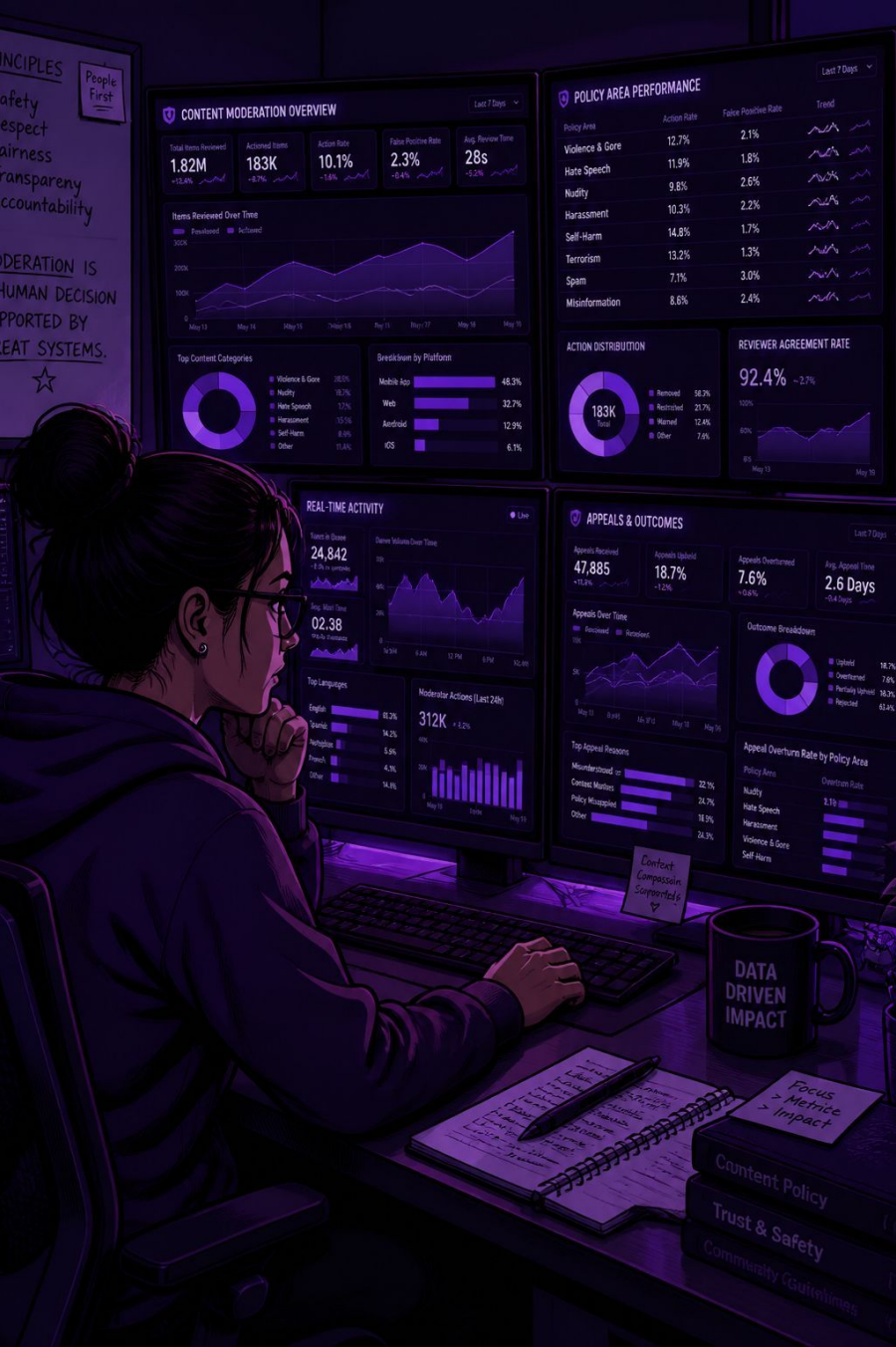




Content Moderation Analytics

An end-to-end data pipeline transforming **1,000,844 raw moderation tickets** into a decision-ready Power BI dashboard — covering data collection, Python cleaning, SQL Server modeling, and interactive reporting.

TRUST & SAFETY

SOURABH YADAV

SQL · POWER BI · PYTHON



Headline Numbers

12-month reporting window: **Aug 2025 – Jul 2026** · 6 violation categories · 5 regions · 4 shifts

1M+

Moderation Tickets

501 moderators, 26 team leads

72%

Human Reviewed

28% auto-actioned by AI

83.4%

Overall Accuracy

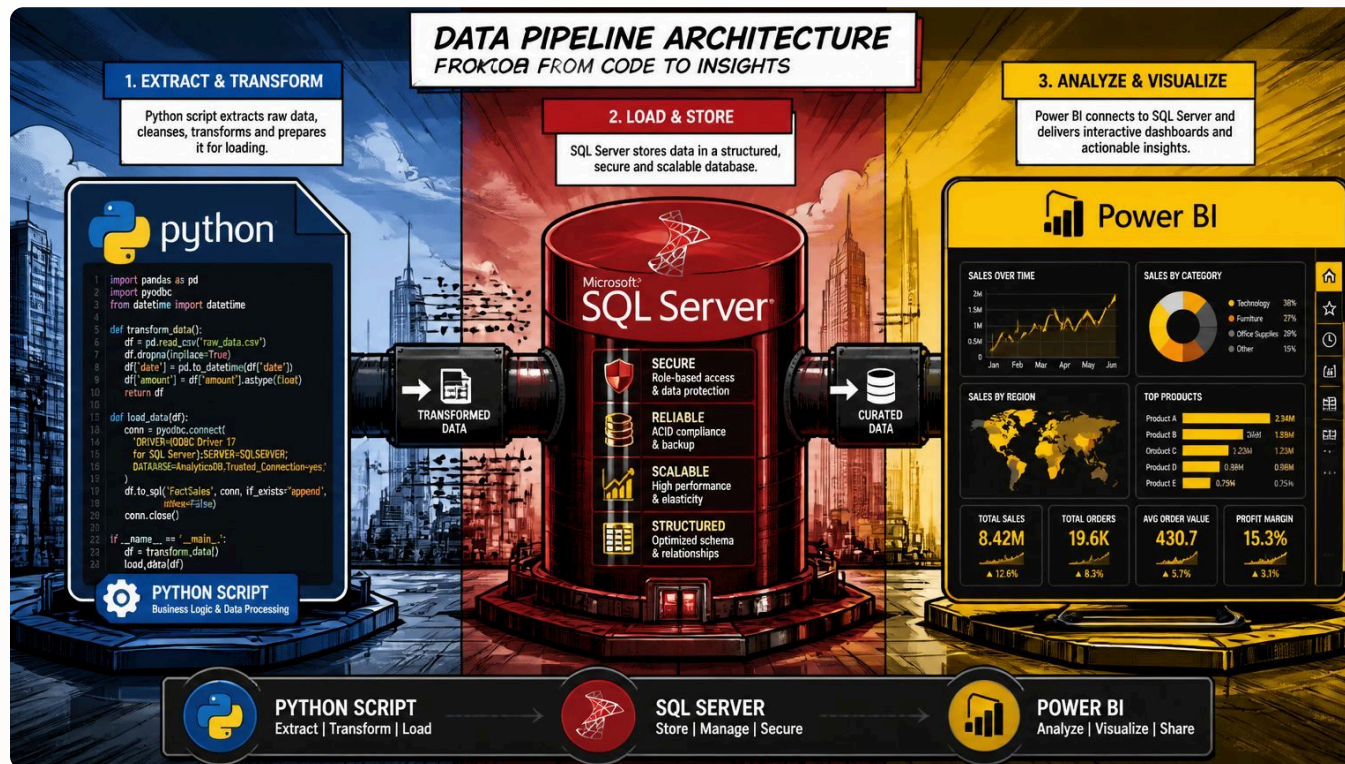
55.7s avg. handling time

34.9%

Hate Speech Overturn

Highest of any category

Project Objective & Tech Stack



Core Business Questions

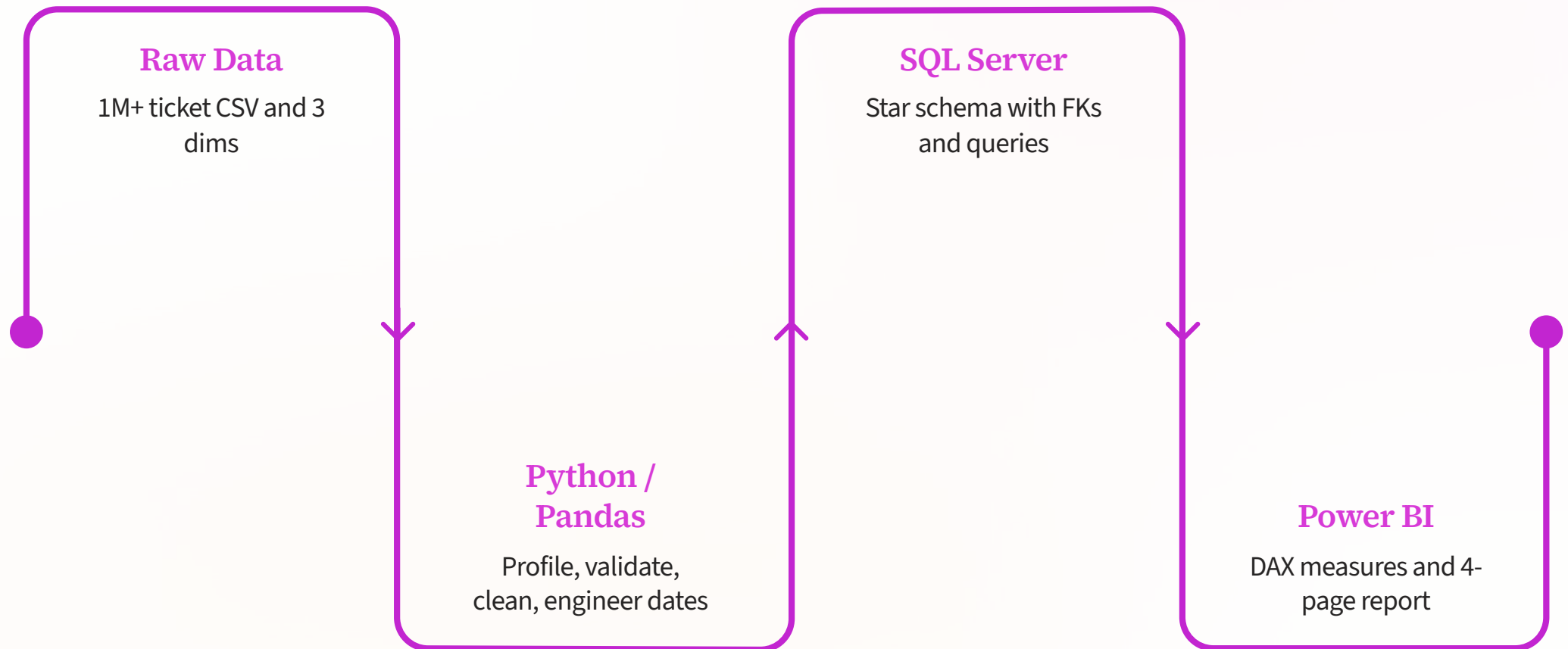
- AI vs. human moderation volume and trends
- Which categories demand the most effort?
- Moderator and team lead quality benchmarks
- Where are appeals and overturns concentrated?

Tech Stack

- **Python** — profiling, cleaning, feature engineering
- **SQL Server** — star schema, T-SQL queries
- **Power BI** — 4-page interactive dashboard
- **GitHub / Notion** — version control & documentation

Data Pipeline Architecture

Each ticket's journey from raw log to dashboard insight.



Four source files: **content_moderation_1M.csv** (fact log) plus **dim_moderator**, **dim_category**, and **dim_team_lead** — mirroring real case-management system exports.

Data Profiling & Cleaning

Profiling Findings

- No duplicate ticket_id or content_id — one row = one decision
- **16.9% missing region** — traced to 117 moderator IDs (source-system gap)
- qa_result & overturn_status nulls validated as correct business logic

Cleaning Actions

- Timestamps converted to datetime
- Missing regions labeled "**Unknown**" — preserving all 169K rows
- Whitespace stripped; no duplicates to remove



Star Schema & SQL Server Modeling



Star Schema Design

- **fact_content_moderation** — one row per ticket decision
- **dim_moderator** → 501 moderators
- **dim_team_lead** → 26 team leads
- **dim_category** → 6 violation categories
- **dim_date** — time-intelligence for Power BI

SQL Implementation

- Row counts validated: 1,000,844 fact rows confirmed
- FK constraints added on **moderator_id**, **team_lead_id**, **category_id**
- 4 indexes on timestamp, category, moderator, region for query performance

Power BI Dashboard — 4 Pages

Executive Summary

6 KPI cards · Cases Over Time by Moderator Type · Severity split · Category workload. Spam/Scam leads at 34.94% of volume.

Moderator Performance

Top 15 moderators · Accuracy vs. AHT scatter · Shift breakdown (Night 34.79% / Evening 32.8% / Morning 32.41%) · Team lead accuracy ranking.

Quality, Risk & Policy

Accuracy by category · Appeal & Overturn table · Action distribution. Hate Speech overturn rate of 34.9% — double every other category.

Moderator Drill-Through

1:1 performance view per moderator — monthly trend, accuracy by category, severity breakdown. Built for team lead coaching conversations.

Executive Overview

Real-time performance overview across all key business dimensions



Revenue Trend



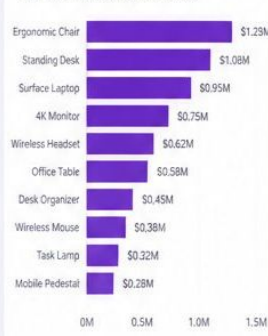
Revenue by Region



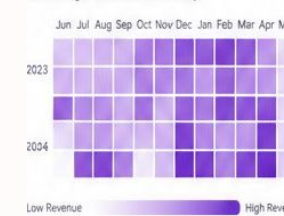
Revenue by Category



Top 10 Products by Revenue



Monthly Revenue Heatmap



Revenue by Channel

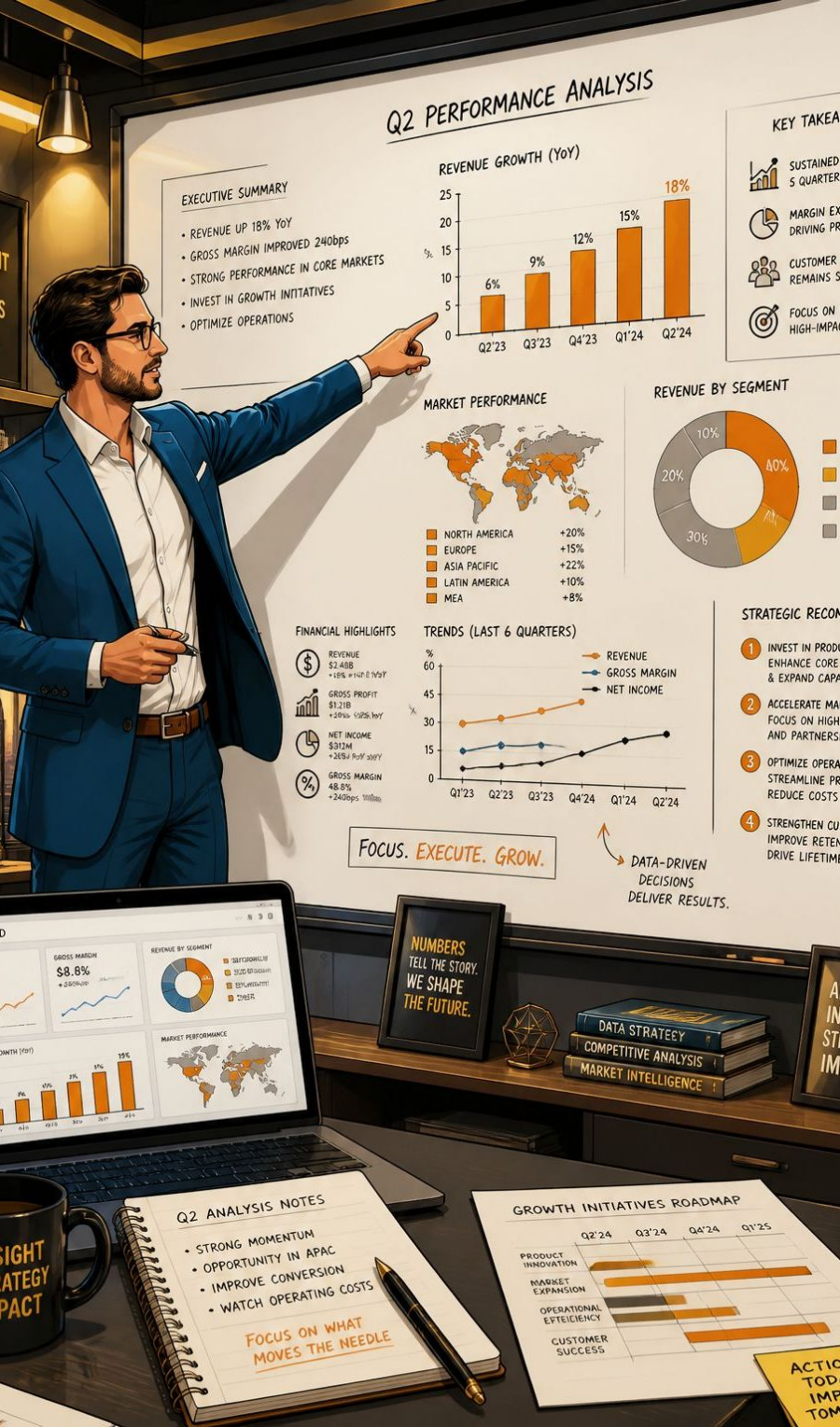


New vs Returning Customers



Performance Summary by Region

Region	Revenue (\$)	Revenue vs PY	Gross Profit (\$)	Gross Profit vs PY	Orders	Orders vs PY	Customers
North America	\$10,450,000	↑ 16.2%	\$4,350,000	↑ 19.3%	7,842	↑ 13.1%	5,642
Europe	\$6,350,000	↑ 14.8%	\$2,450,000	↑ 17.1%	4,512	↑ 11.2%	2,942
Asia Pacific	\$4,680,000	↑ 13.9%	\$1,820,000	↑ 16.5%	3,256	↑ 10.3%	2,142
South America	\$1,820,000	↑ 12.1%	\$680,000	↑ 13.2%	1,352	↑ 9.4%	812
Middle East & Africa	\$1,230,000	↑ 10.7%	\$430,000	↑ 11.8%	780	↑ 8.2%	482
Total	\$24,530,000	↑ 15.6%	\$9,850,000	↑ 18.3%	18,742	↑ 12.4%	12,042



Key Insights & Recommendations

→ Hate Speech: Policy-Consistency Risk

34.9% overturn — double every category. Run calibration sessions and expand re-audit sample.

→ Abigail Shaffer's Team: Coaching Priority

59.06% accuracy — 20+ points below the next-lowest team. Immediate audit and coaching needed.

→ Expand AI Automation

AI handles 27.9% of volume. Spam/Scam is a strong next candidate, freeing humans for Violent Extremism and Sexual Content.

→ Increase Audit Coverage

Only 1.44% of cases audited — the 83.4% accuracy figure rests on a thin sample. Expand especially for Hate Speech and Violent Extremism.